

Edexcel A-Level Mathematics

Paper 3 (9MA0/03) - Statistics & Mechanics

Last-Minute Rescue Sheet | June 2026

Paper 3 splits down the middle: 50 marks Statistics, 50 marks Mechanics. Different skills, different traps, same marking system. This sheet is built from Pearson's examiner reports (2019–2024), published mark schemes, and the 9MA0 specification. Everything here exists to answer one question: where are you most likely to drop marks, and how do you stop it?

Two things to nail down right now: (1) SUVAT equations are in the formula booklet. $F = ma$ is not. (2) The word "sufficient" will earn you marks. The word "enough" won't. Details like these are the difference between a B and an A.

1. Formula Triage - Statistics

The booklet gives you the complex derivation formulas. It does NOT give you the practical stuff you need actually to apply them. The right column is non-negotiable.

IN THE BOOKLET (don't memorise)	NOT IN BOOKLET (must know cold)
S_{xx} , S_{yy} , S_{xy} sum-of-squares formulas	Outlier boundaries: $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$ (or $\mu \pm 2\sigma$)
Standard deviation and variance formulas	Frequency density = Frequency / Class width
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	Linear interpolation: Lower boundary + (target - cum.freq) / class freq $\times$ class width
Conditional probability: $P(A B) = P(A \cap B) / P(B)$	Data coding: If $y = (x-a)/b$ then $\sigma_y = \sigma_x/b$ (addition/subtraction does NOT affect s.d.)
Binomial probability formula	Conditions for Binomial: Fixed n , constant p , independent, two outcomes
Binomial mean (np) and variance ($np(1-p)$)	Normal approx to Binomial: $\mu = np$, $\sigma = \sqrt{np(1-p)}$ Continuity correction ± 0.5 REQUIRED
Normal standardisation: $Z = (X - \mu) / \sigma$	Hypothesis test structure: See Section 3

2. Formula Triage - Mechanics

Students assume everything in Mechanics is in the booklet. Wrong. SUVAT is given. Almost everything else isn't. The right column is where marks live.

IN THE BOOKLET	NOT IN BOOKLET (must memorise)
SUVAT equations (scalar): $v = u + at$ $s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$ $s = \frac{1}{2}(u+v)t$	Newton's Second Law: $F = ma$ (Yes, really. It's not in the booklet.)
SUVAT equations (vector form)	Weight: $W = mg$ Always use $g = 9.8$ unless told otherwise
-	Friction: $F = \mu R$ (at the point of slipping) $F \leq \mu R$ (general case)
-	Moments: Force $\times$ perpendicular distance Equilibrium: sum of clockwise = sum of anticlockwise
-	Variable acceleration (calculus): $v = ds/dt$, $a = dv/dt$ $s = \int v \, dt$, $v = \int a \, dt$

-	Resolving on inclined planes: Parallel to slope = $mg \sin \theta$ Perpendicular to slope = $mg \cos \theta$
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3. The Hypothesis Testing Blueprint

Same exam, same trap, every year. Students know the maths but lose marks because they skip a step or use the wrong word. Here's the exact structure Edexcel's mark scheme rewards.

Worked Example

Historically, 15% of seeds fail to germinate. A new fertiliser is tested on 40 seeds. Only 2 fail. Test at 5% significance whether the fertiliser has reduced the failure rate.

Step 1 - Define the Model	Let X = number of seeds failing to germinate. $X \sim B(40, 0.15)$	B1 mark. Write the distribution with actual numbers from the question.
Step 2 - State Hypotheses	$H_0: p = 0.15$ $H_1: p < 0.15$	B1 mark. Must use p (not x , not μ). One-tailed because we're testing for reduction.
Step 3 - Calculate P-Value	$P(X \leq 2) = 0.0486$	M1 mark. Use cumulative binomial on calculator.
Step 4 - Compare	$0.0486 < 0.05$	M1 mark. Write the inequality. Don't just say 'it's significant'.
Step 5 - Two-Part Conclusion	Reject H_0 . There is sufficient evidence to suggest the failure rate of seeds has reduced.	A1 mark. Must say 'sufficient' not 'enough'. Must say 'suggest' not 'prove'. Must reference actual scenario.

SCORES MARKS	LOSES MARKS
"sufficient evidence"	"enough evidence"
"suggest that..."	"show" or "prove that..."
"reject H_0 " / "do not reject H_0 "	"accept H_1 " or only mentioning H_1
Conclusion uses exact wording from the question	Generic conclusion with no context variables

4. Large Data Set - Quick-Reference

You don't need to memorise the numbers. You need to know the quirks. Every item below has appeared in exam questions, and every one has tripped students up.

Feature	What It Means	The Exam Trap
Locations	UK: Camborne, Heathrow, Hurn, Leeming, Leuchars International: Beijing, Jacksonville, Perth	Perth is in Australia (Southern Hemisphere). May–Oct is their winter . Temperatures fall while UK rises.
"tr" (Trace)	Rainfall less than 0.05 mm	When calculating a mean or total, treat tr as 0 . Not 0.05. Zero.
Oktas	Cloud cover in eighths of the sky. Values: 0–8.	This is discrete data. A mean of 4.5 is fine, a single reading of 4.5 is impossible.
Missing data (n/a)	Some days have no reading recorded.	If 2 days out of 30 are n/a, divide totals by 28 , not 30.
Wind direction	Measured in degrees and compass points.	This is qualitative . You can find the mode but NOT the mean or median.
Humidity	Relative humidity as a percentage.	Above 95% is associated with mist and fog.

Correlation vs. causation: Never write "causes". Write "there appears to be a relationship between" or "is associated with". Examiners dock marks for causal language. Every single time.

5. Mechanics Problem-Solving Frameworks

Connected Particles & Pulleys

Examiners expect simultaneous equations. Don't try to treat the whole system as one body when friction is involved - it doesn't work and you'll lose method marks.

Step	What to Do	Why It Matters
1	Draw separate force diagrams for each particle	Combining them causes sign errors.
2	Apply $F = ma$ to each particle individually	M1 mark for each correct equation.
3	T is the same on both sides (smooth pulley)	This lets you solve simultaneously.
4	Acceleration magnitude is the same for both particles	Connected by an inextensible string.
5	Add the two equations to eliminate T, then solve for a	Then sub back to find T.

Moments - The Twist Variations

Scenario	What It Actually Means	Your Move
"Uniform rod"	Weight acts at the exact midpoint.	Mark the midpoint on your diagram. Put mg there.
"Non-uniform rod"	Weight acts at an unknown point.	Call the distance x. Solve for it.
"On the point of tilting"	Normal reaction at one support = 0.	Set $R = 0$ at the support about to lift.
"Light rod/beam"	The rod has no weight. Don't add mg.	Only external forces create moments.

Pro tip: Take moments about the point where an unknown force acts. That eliminates it from the equation and saves you an entire step.

Variable Acceleration - The One Trap That Costs 3+ Marks

If acceleration is a function of time (e.g. $a = 3t^2 - 4$), you can't use SUVAT. Full stop. You need calculus: integrate a to get v, integrate v to get s.

The trap that catches roughly half of all students: if a question asks for total distance (not displacement), you must check whether $v = 0$ at any point in the interval. If the particle changes direction, integrate each section separately and add the absolute values. Integrating straight across gives displacement - which could be zero even though the particle moved.

And don't forget the constant of integration. Every time you integrate, write + C. Then use the initial conditions to find it. Missing + C costs 1–2 marks automatically.

6. SUVAT Decision Tree

For constant acceleration only. List s, u, v, a, t. Fill in the three you know. The variable you don't care about tells you which equation to use.

Variable You DON'T Need	Equation to Use	Common Scenario
s	$v = u + at$	Finding final velocity, don't care about distance
v	$s = ut + \frac{1}{2}at^2$	Finding distance when final velocity isn't given
t	$v^2 = u^2 + 2as$	Maximum height ($v = 0$), final speed after a distance
a	$s = \frac{1}{2}(u + v)t$	When acceleration isn't mentioned

Quick checks: At maximum height, $v = 0$. At the starting point on return, $s = 0$. Object dropped from rest: $u = 0$. These initial conditions are free marks if you spot them.

7. Examiner Red Flags & M1/A1 Patterns

Here's how Edexcel's marking actually works in Mechanics. Understanding this turns partial answers into partial credit.

Situation	What Gets M1	What Gets M0	Common A1 Loss
Resolving forces	Correct number of terms, all forces included, sin/cos attempted	Omitting mass from the equation = instant M0	Swapping sin and cos on incline
Taking moments	Force $\times$ distance for every force, all resolved	Omitting a distance from a moment = M0	Wrong perpendicular distance
Applying $F = ma$	Equation with all relevant forces, dimensionally correct	Completely wrong approach or missing key force	Sign error (inconsistent positive direction)
Any calculation	Correct setup with numbers substituted	No method shown at all	Using $g = 9.81$ instead of 9.8 (loses accuracy mark)

The $g = 9.8$ rule: Edexcel uses $g = 9.8$ unless the question says otherwise. Using 9.81 means your final answer won't match the mark scheme. If you used 9.8 , give your answer to 2 or 3 significant figures.

8. The Last 30 Minutes - M1 Mark Grabbing

Time's nearly up and you've got questions left. Stop trying to finish one perfectly. Spread your remaining time across every unanswered question and collect method marks. A single M1 from four different questions beats a complete answer on one.

Mechanics - Quick Wins

Action	Time	Marks
Draw and label a force diagram: weight (mg) down, R perpendicular, friction opposing motion	1 min	1 (B1)
Object on a slope? Write: parallel = $mg \sin \theta$, perpendicular: $R = mg \cos \theta$	30 sec	2 (M1+M1)
See a ladder or plank? Write "Moments about A:" and list Force $\times$ Distance for every arrow	1 min	1 (M1)
See $v = 3t^2 + 2$ and question asks for distance? Write: $s = \int (3t^2 + 2) dt$	15 sec	1 (M1)

Statistics - Quick Wins

Action	Time	Marks
Write H_0 and H_1 with correct parameter (p for proportion, μ for mean, ρ for correlation)	15 sec	1–2 (B1)
Define the distribution: $X \sim B(n, p)$ or $X \sim N(\mu, \sigma^2)$ with actual numbers	10 sec	1 (B1)
Dense probability question? Draw the Venn diagram or tree and fill in given numbers	1 min	1–2 (M1)
Write "Let p be the proportion of [context from question]"	10 sec	1 (B1)

One last thing. Force diagrams aren't optional - they're marking points. Hypothesis test conclusions aren't creative writing - they're a formula. Use $g = 9.8$. Say "sufficient", not "enough". These details are the difference between knowing the maths and actually scoring for it.

Built by StudyHours.com from Pearson Edexcel examiner reports (2019–2024), official 9MA0 mark schemes, and the Edexcel A-Level Mathematics specification.